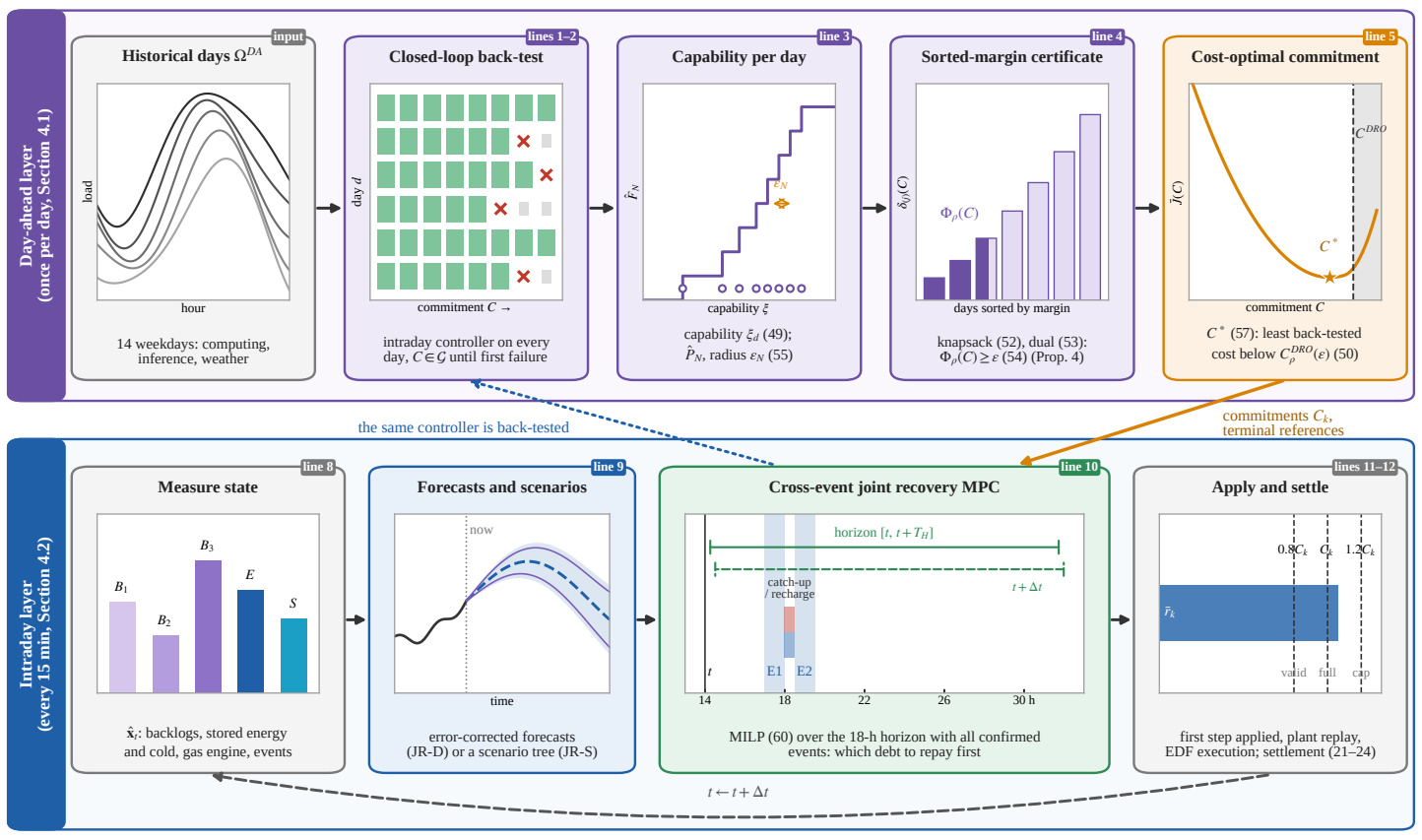

Day-ahead layer
(once per day, Section 4.1)
input
Historical days Ω^{DA}
load
hour
14 weekdays: computing, inference, weather
lines 1–2
Closed-loop back-test
day d
commitment C →
intraday controller on every day, $C \in \mathcal{G}$ until first failure
line 3
Capability per day
$\hat{P}_N$
ε_N
capability ξ
capability ξ_d (49); $\hat{P}_N$, radius ε_N (55)
line 4
Sorted-margin certificate
$\delta_{(j)}(C)$
$\Phi_\rho(C)$
days sorted by margin
knapsack (52), dual (53): $\Phi_\rho(C) \geq \varepsilon$ (54) (Prop. 4)
line 5
Cost-optimal commitment
$\bar{J}(C)$
C^{DRO}
C^*
commitment C
C^* (57): least back-tested cost below $C_\rho^{DRO}(\varepsilon)$ (50)
the same controller is back-tested
commitments C_k, terminal references
Intraday layer
(every 15 min, Section 4.2)
line 8
Measure state
B_1
B_2
B_3
E
S
$\hat{\mathbf{x}}_t$: backlogs, stored energy and cold, gas engine, events
line 9
Forecasts and scenarios
now
time
error-corrected forecasts (JR-D) or a scenario tree (JR-S)
line 10
Cross-event joint recovery MPC
horizon $[t, t+T_H]$
$t+\Delta t$
catch-up / recharge
t
E1
E2
14
18
22
26
30 h
MILP (60) over the 18-h horizon with all confirmed events: which debt to repay first
lines 11–12
Apply and settle
$0.8C_k$
C_k
$1.2C_k$
$\bar{r}_k$
valid
full
cap
first step applied, plant replay, EDF execution; settlement (21–24)
$t \leftarrow t + \Delta t$